



Original software publication

AIDSII: An AI-based digital system for intelligent irrigation

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ABSTRACT

In recent years, precision agriculture has gained significant attention as a result of the rising demand for food and water by the world population and the increasing impact of climate change. This highlights the need for innovative solutions to address this critical challenge. Farmers are facing a shortage of water and fertile land, which demands a new approach to their work. Intelligent irrigation is crucial to improve yields and maximize the use of resources. Various machine learning-based irrigation models have been proposed to minimize water wastage. This paper introduces AIDSII, an AI-powered digital application that leverages IoT-based precision agriculture and CNN-LSTM models. It offers a comprehensive feedback system through mobile and web technologies, enabling farmers to automate, optimize, and streamline their irrigation processes.

Code metadata

Current code version

Permanent link to code/repository used for this code version

Code versioning system used

Software code languages, tools and services used

Compilation requirements, operating environments and dependencies

If available, link to developer documentation

Support email for questions

V1

<https://github.com/SoftwareImpacts/SIMPAC-2023-296>

Git (used to track changes and collaborate on code development)

Java, HTML, CSS, JS, Spring Boot, Spring Security, Thymeleaf, Bootstrap, JQuery, Android, C, MySQL

Java 9 or higher, Java compiler version, Spring Framework, Maven, Java Development Kit

https://github.com/raouhi/smart_wateringmaster/README.mdraouhi@gmail.comlachgar.m@gmail.com

1. Introduction

The Internet of Things (IoT) refers to the growing network of physical devices, vehicles, home appliances, and other objects that are embedded with electronics, software, sensors, and connectivity which enables these objects to collect and exchange data [1]. IoT technology allows these objects to be connected and controlled remotely, enabling a new level of automation and control in various industries and aspects of daily life. In the context of IoT, devices and objects can be equipped with sensors that collect data on things like temperature, humidity, and motion, and then send that data over the internet to be analyzed and used for various purposes. This data can be used to trigger automated actions, such as turning on a light when someone enters a room, or

to provide valuable insights for businesses and individuals, such as tracking the usage patterns of a piece of equipment [2–4].

The potential applications of IoT are vast and span across multiple industries, including agriculture, healthcare, manufacturing, transportation, and energy, among others. In agriculture, for example, IoT can be used to monitor soil moisture levels, track the health of crops, and manage irrigation systems. In the healthcare industry, IoT devices can be used to monitor vital signs and provide real-time health data to medical professionals. Overall, the IoT is transforming the way that people live, work, and interact with the world around them, providing new levels of automation and control that were previously not possible [5–9].

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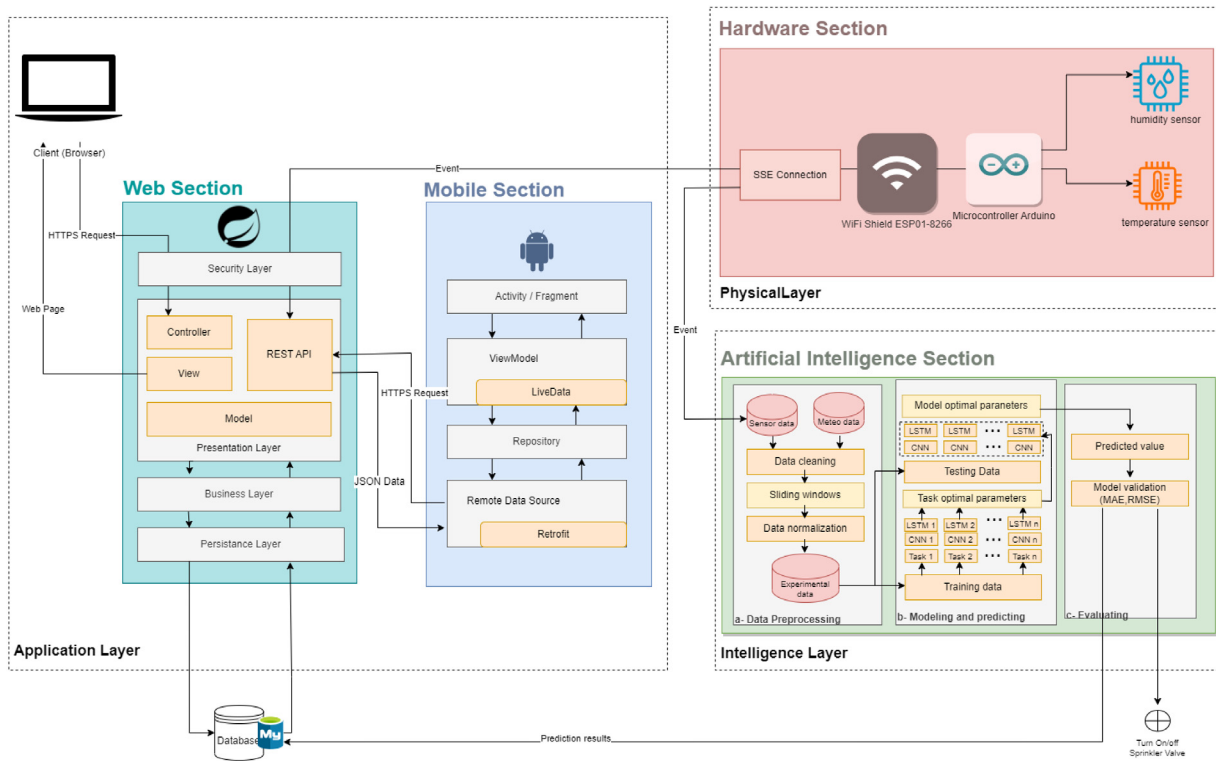


Fig. 1. High level architecture: AI-based digital application for intelligent irrigation.

Water is one of the most essential resources for all forms of life, as well as for human activities such as agriculture, industry, and daily life. It is a finite resource, and access to clean, safe water is becoming increasingly important as populations grow and demands on water resources increase. Intelligent irrigation refers to the use of advanced technologies and techniques to manage and optimize the watering of crops or plants [10]. This can include using sensors to monitor soil moisture levels, weather forecasts to predict precipitation, and automatic control systems to regulate the amount and timing of water applied to the plants. The goal of intelligent irrigation is to maximize crop yields while minimizing water waste, energy use, and other resources. Some of the benefits of intelligent irrigation systems include improved crop health, reduced water usage, and cost savings on energy and water bills. By using real-time data to inform irrigation decisions, these systems can also help reduce the risk of water scarcity and environmental degradation associated with traditional irrigation methods [11]. There are many different types of intelligent irrigation systems available, ranging from simple timers and controllers to more advanced systems that use artificial intelligence to make irrigation decisions. Regardless of the type of system used, the key to successful intelligent irrigation is to carefully analyze the unique needs of each individual crop and to adjust the system accordingly [12].

In this paper, we delve into the implementation details of this software and examine its tangible impact in the real world. Our primary focus centers around elucidating the design concepts of the software architecture and providing intricate insights into the engineering considerations involved and exploring the profound effects it has on agricultural practices.

2. System description

2.1. Software architecture

This section provides a comprehensive description of the software, including its components and layers: Application layer, Intelligence layer, and Hardware layer. The architecture of the AI-based digital

system for intelligent irrigation, which encompasses both hardware and Artificial Intelligence (AI) components, is depicted in Fig. 1.

3. Impact overview

The AIDSII system primarily focuses on enhancing irrigation practices. However, the implemented source code can be extended to support various other applications [13]. The main concept of this software is to optimize water distribution by targeting specific areas and employing advanced techniques for efficient irrigation. The versatility of our system allows for the implementation of additional features, such as soil moisture monitoring and crop yield prediction. Users can then retrain our model accordingly. The resulting model can be tested using specific irrigation zones, such as fields or individual plants, to generate optimized irrigation plans. Therefore, it can also be used to support farmers in making informed decisions and maximizing crop productivity. This would facilitate better understanding and discussion of irrigation practices among stakeholders.

In continuation of the work undertaken for the implementation of an integrated system for a smart farm [14], specifically for olive crops, the AI-powered irrigation system revolutionizes farming practices by providing farmers with real-time data analysis and predictive insights to optimize crop production and conserve water resources. Through advanced algorithms, the app enables precise water management by considering factors such as soil moisture levels, weather conditions, and crop requirements, resulting in optimal irrigation schedules and minimized water wastage. Additionally, the app offers smart water allocation, proactive monitoring with alerts, and historical analysis for informed decision-making. By empowering farmers with intelligent tools, the app facilitates sustainable agriculture practices, maximizing crop yields while minimizing the environmental impact of irrigation.

4. Limitations and future work

Nonetheless, it is essential to recognize specific limitations within the AIDSII system. The effectiveness of the AI model relies heavily on

the quality and precision of sensor input data; potential inaccuracies could result in suboptimal irrigation choices. Furthermore, despite providing room for additional features, the system's implementation might pose a challenge for users who are less familiar with AI. To address these concerns, future research could concentrate on refining AI algorithms to adeptly handle noisy or incomplete data, enhancing the user interface for intuitive ease-of-use and improved accessibility, and exploring the integration of AI-powered pest and disease prediction models for a comprehensive agricultural management solution. By effectively addressing these constraints and remaining committed to these anticipated advancements, the AIDSII system holds the potential to evolve into an even more resilient and indispensable tool within the precision agriculture domain.

5. Conclusion

In this study, a smart irrigation system based on the CNN-LSTM model is proposed. It aims to address the key requirements of agricultural irrigation, such as water supply and irrigation timing, by controlling the irrigation scheduling function. The proposed system was validated through tests utilizing various environmental factors like temperature and humidity collected from sensors. The AIDSII, an AI-based digital system for intelligent irrigation, has been developed using the CNN-LSTM model. This application is well-suited for classifying, processing, and making predictions from data obtained from IoT sensors. In future research, the aim is to design predictive irrigation schedules that can predict rainfall depth and soil moisture, resulting in better water storage by enhancing rainfall depth.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.simpa.2023.100574>.

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